

AI-Powered Customer Churn Intelligence Platform

Project Report

Submitted By

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1. Introduction

Customer churn is one of the biggest challenges faced by banks and financial institutions. Customer churn refers to customers leaving a company or discontinuing its services. Predicting churn helps organizations identify at-risk customers and take preventive actions.

This project develops an AI-powered Customer Churn Intelligence Platform that uses Machine Learning to predict whether a customer is likely to leave the bank. The solution includes an interactive Streamlit dashboard, data analytics, visualizations, and FastAPI integration for real-time predictions.

2. Objectives

- Predict customer churn using Machine Learning.
 - Analyze customer behavior and banking patterns.
 - Build an interactive dashboard for business insights.
 - Identify high-risk customers.
 - Support customer retention strategies.
 - Deploy prediction services using FastAPI.
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3. Dataset Description

Dataset Used: European Bank Customer Dataset

Features

- CreditScore
- Age
- Tenure
- Balance
- NumOfProducts
- HasCrCard
- IsActiveMember
- EstimatedSalary
- Geography
- Gender

Target Variable

Exited

- 0 = Customer Stays
 - 1 = Customer Churns
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4. Data Preprocessing

The following preprocessing steps were performed:

- Handling missing values
 - Data cleaning
 - Feature engineering
 - Encoding categorical variables
 - Feature scaling where required
 - Data splitting into training and testing sets
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5. Machine Learning Model

Algorithm Used

Random Forest Classifier

Reasons for selecting Random Forest:

- High prediction accuracy
- Handles large datasets effectively
- Reduces overfitting
- Works well with both numerical and categorical data

Model Workflow

1. Load Dataset
2. Data Cleaning
3. Feature Engineering
4. Train-Test Split
5. Model Training
6. Model Evaluation
7. Model Saving using Joblib

Model File:

churn_model.pkl

6. Dashboard Development

The dashboard was developed using Streamlit.

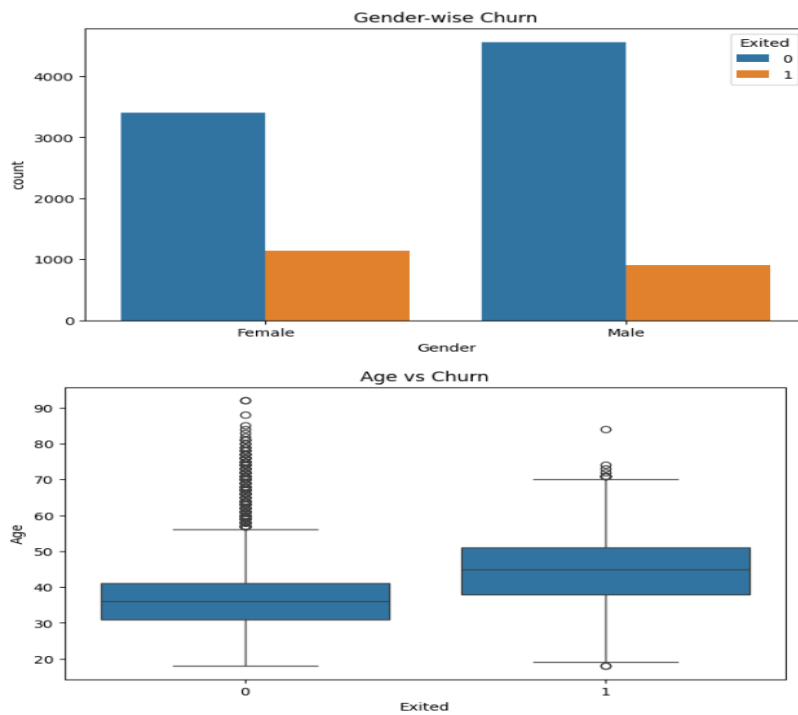
Key Features

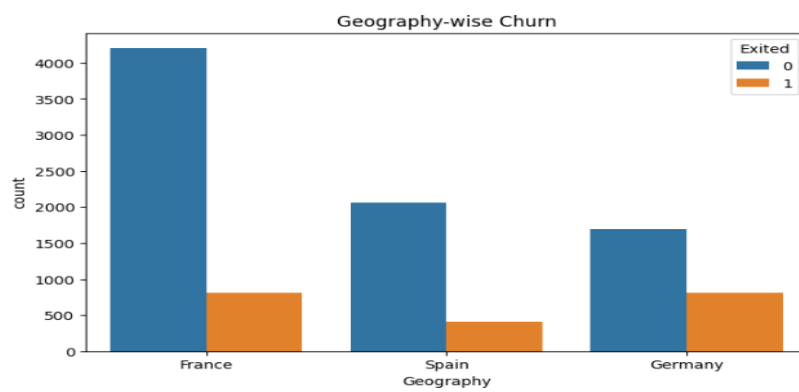
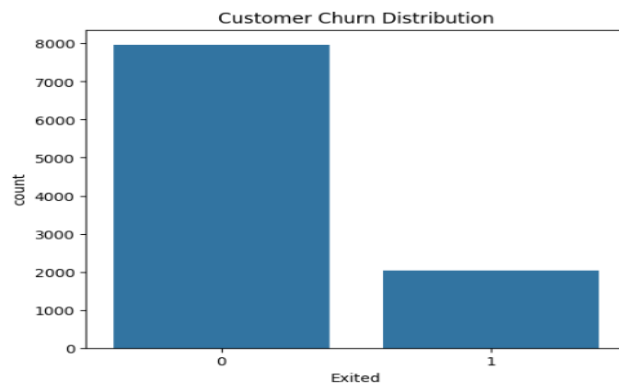
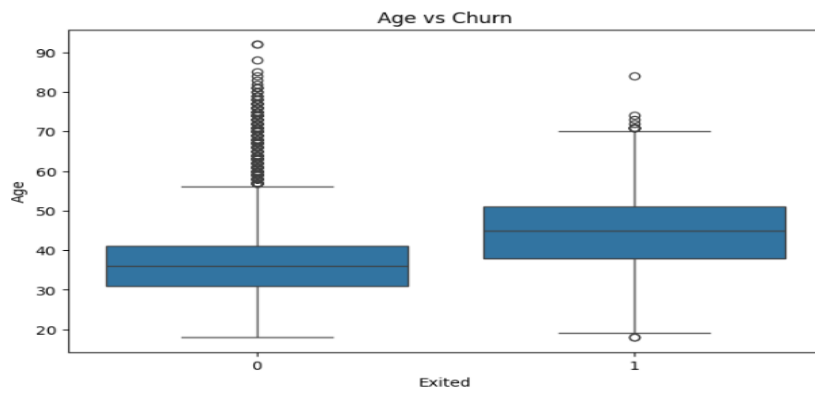
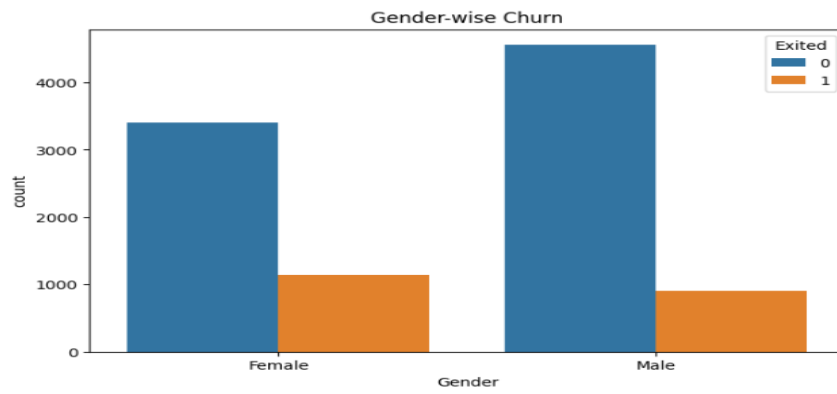
Executive Dashboard

- Total Customers
- Churn Rate
- Average Balance
- High Value Customers
- Revenue Risk Analysis

Visualizations

- Customer Churn Distribution
- Age vs Churn Analysis
- Geography-wise Churn Rate
- Balance vs Churn Analysis
- High Value Customer Analysis





AI Prediction Module

Users can enter customer information and receive:

- Churn Prediction

The interface displays a 'Predict Churn' button at the top. Below it, the 'Churn Probability' is shown as 50.00%, and the 'Prediction' result is 'STAY'. A yellow bar indicates the customer is a 'MEDIUM RISK CUSTOMER', and a green bar states 'Customer is likely to STAY'. A 'Download Prediction Report' button is at the bottom.

Predict Churn

Churn Probability
50.00%

Prediction
STAY

MEDIUM RISK CUSTOMER

Customer is likely to STAY

Download Prediction Report

- Churn Probability

The dashboard includes a 'Filters' sidebar with 'Geography' (All), 'Gender' (All), 'Internship Project', and 'AI Banking Analytics Dashboard'. The main area is titled 'AI Customer Churn Predictor' and contains input fields for 'Credit Score' (650), 'Age' (35), 'Tenure' (5), 'Balance' (50000.03), 'Number of Products' (2), 'Has Credit Card' (0), 'Is Active Member' (0), and 'Estimated Salary' (50000.00). A 'Predict Churn' button is present. The results section shows a 'Churn Probability' of 50.00%, a 'Prediction' of 'STAY', a 'MEDIUM RISK CUSTOMER' status, and a confirmation that the 'Customer is likely to STAY'. A 'Download Prediction Report' button is also available.

Filters

Geography
All

Gender
All

Internship Project

AI Banking Analytics Dashboard

AI Customer Churn Predictor

Credit Score: 650

Age: 35

Tenure: 5

Balance: 50000.03

Number of Products: 2

Has Credit Card: 0

Is Active Member: 0

Estimated Salary: 50000.00

Predict Churn

Churn Probability
50.00%

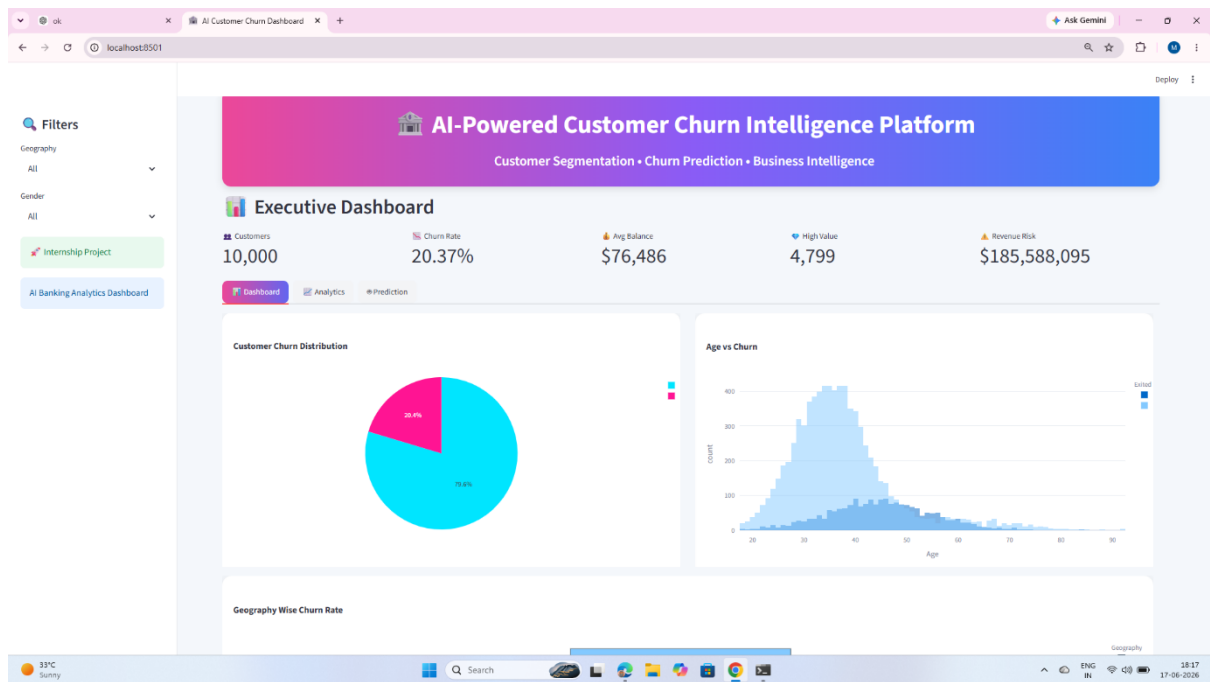
Prediction
STAY

MEDIUM RISK CUSTOMER

Customer is likely to STAY

Download Prediction Report

Output



7. FastAPI Integration

FastAPI was used to create prediction endpoints.

Endpoints

GET /

Returns API status.

POST /predict

Returns customer churn prediction based on input features.

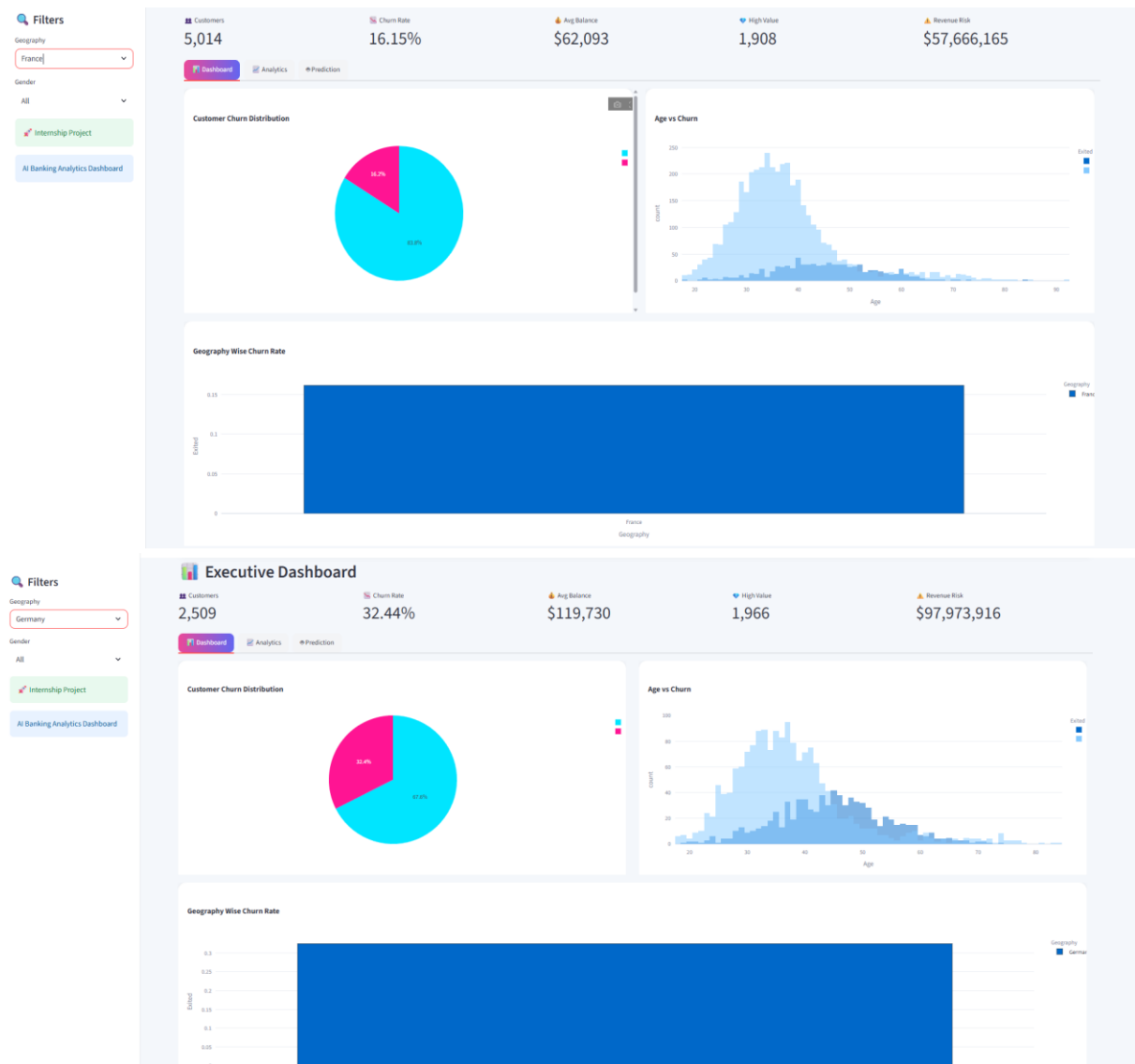
Benefits

- Real-time predictions
- Easy integration with applications
- Scalable deployment architecture

8. Results and Findings

Key insights obtained from analysis:

- Germany shows higher customer churn compared to other regions.
- Inactive customers are more likely to leave the bank.
- Older customers have a higher churn probability.
- High-value customers require retention strategies.
- Customer activity significantly impacts churn behavior.



9. Business Impact

The developed system can help banks:

- Reduce customer churn.
 - Improve customer retention.
 - Increase profitability.
 - Identify high-risk customers early.
 - Support data-driven decision making.
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10. Technologies Used

- Python
- Pandas
- NumPy
- Scikit-Learn
- Plotly
- Streamlit
- FastAPI
- Joblib

11. Future Scope

Future improvements may include:

- Deployment on cloud platforms
 - Deep Learning models
 - Automated customer recommendation systems
 - Real-time streaming analytics
 - Customer retention recommendation engine
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12. Conclusion

The AI-Powered Customer Churn Intelligence Platform successfully predicts customer churn and provides valuable business insights through interactive dashboards and visual analytics. The project demonstrates practical applications of Machine Learning, Data Analytics, Data Visualization, and API Deployment in solving real-world business problems.

References

1. Scikit-Learn Documentation
 2. Streamlit Documentation
 3. FastAPI Documentation
 4. Plotly Documentation
 5. European Bank Customer Dataset
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